Nadiminti Venkata Sai Tej

9490236117 | mrtej117@gmail.com | LinkedIn | GitHub | Vizianagaram, Andhra Pradesh

Professional Summary

Second-year B.Tech Computer Science and Engineering student specializing in Artificial Intelligence and Machine Learning. Hands-on experience in Python, PyTorch, PySpark, deep learning, computer vision, and predictive analytics. Developed image captioning systems, distributed machine learning pipelines, and real-time computer vision applications. Interested in solving real-world problems through AI, Machine Learning, and Data Science.

Education

Centurion University of Technology and Management — **CGPA: 8.67/10**
B.Tech, Computer Science and Engineering (AI & ML) — *2025 – Present*

Aditya Junior College — **89.2%**
Intermediate (MPC) — *2023 – 2025*

Narayana Group of Schools — **88.8%**
Secondary School Certificate (SSC) — *2022 – 2023*

Technical Skills

Programming & Operating Systems: Python, Java, C, Linux
AI & Deep Learning: Machine Learning, Deep Learning, PyTorch, CNN, ResNet-50, LSTM, Computer Vision, NLP
Data Science: EDA, Statistical Analysis, Data Preprocessing, Feature Engineering, NumPy, Pandas, Matplotlib
Big Data: PySpark, Spark MLlib, Decision Trees
Cloud & AI: Microsoft Azure, Azure AI Services, Azure AI Foundry, Azure Machine Learning, Generative AI
Databases & Tools: PostgreSQL, MS SQL, Git, GitHub, VS Code, Jupyter Notebook, Google Colab, Anaconda, OpenCV, MediaPipe

Projects

Vision2Text: Image Captioning using CNN-LSTM — *PyTorch, ResNet-50, LSTM* — GitHub

- Developed an image captioning system using a pretrained ResNet-50 CNN encoder and LSTM decoder to generate meaningful natural-language descriptions from images.
- Implemented image preprocessing, vocabulary construction, tokenization, feature extraction, and sequence generation using PyTorch.
- Trained for 10 epochs on MS COCO 2017 and evaluated on 5,000 validation images, achieving BLEU-1 of 69.85% and BLEU-4 of 27.40%.

Customer Churn Prediction using PySpark — *PySpark, Spark MLlib* — GitHub

- Built a distributed machine learning pipeline using PySpark to analyze 7,043 telecom customer records and predict customer churn.
- Performed EDA, missing-value imputation, outlier removal, categorical encoding, feature engineering, and scaling.
- Trained and tuned a Spark MLlib Decision Tree classifier, achieving a best test ROC-AUC of 0.7789 and analyzing overfitting and important churn predictors.

ASCII Depth Rain: Real-Time Computer Vision — *Python, OpenCV, MediaPipe* — GitHub

- Developed a real-time webcam application that transforms human face and body silhouettes into dynamic ASCII art using MediaPipe segmentation.
- Implemented brightness-based ASCII mapping, Matrix-style digital rain, multiple color palettes, and interactive rendering controls using OpenCV.

Experience

E&ICT Academy, IIT Guwahati — **Remote**
Artificial Intelligence & Machine Learning Intern — *July 2026 – September 2026*

- Completed an AI/ML internship from July 19 to September 19, 2026, working on deep learning applications, Visual Question Answering (VQA), and video captioning.
- Developed and submitted an Image Captioning project integrating computer vision and natural language processing techniques.

Certifications

- **Microsoft Certified: Azure AI Fundamentals (AI-901)** – Microsoft [Verify]
- **Google AI Professional Certificate** – Google [Verify]
- **AI/ML Internship Certificate** – E&ICT Academy, IIT Guwahati
- **Machine Learning with PySpark: Customer Churn Analysis** – Coursera [Verify]
- **Qiskit Global Summer School 2026 – Quantum Excellence** – IBM [Verify]